HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026

GlucoTwin Bharat

An Indian meal-aware, personalized glucose-excursion forecasting Digital Twin

REACT + FASTAPI + ML

Solo student prototype | Synthetic patients and sensor streams only

The gap: dashboards describe; twins update and forecast

Focus: one localized outcome — potential elevated-glucose excursion within 120 minutes.

TODAY: HISTORICAL MONITORING

- Readings are reviewed after collection
- Meal and activity context may be fragmented
- A static risk score ignores changing state

GLUCOTWIN: DYNAMIC ESTIMATION

- Fuses synthetic EHR + time-series observations
- Updates a per-patient virtual state
- Forecasts multiple horizons
- Tests hypothetical Indian-meal scenarios

A narrowly scoped Digital Twin

A proof-of-concept is more credible when its target, inputs and limits are explicit.

Historical EHR

Demographics, HbA1c, diagnoses, baseline glucose

→

Dynamic stream

CGM, meal events, steps, sleep, recent trend

→

Twin engine

Patient-specific features + updated state

→

Outputs

15/30/60/120-minute trajectory + scenario delta

Key distinction: the twin's forecast updates when observations and scenario inputs change.

Use case: meal-aware glucose excursion estimation

Indian meals serve as scenario labels; carbohydrate grams are explicit, editable estimates.

INPUTS

- Current glucose
- Recent glucose slope
- Estimated meal carbs
- Steps and sleep
- Individual synthetic response

FORECAST

- +15 min
- +30 min
- +60 min
- +120 min
- Threshold-crossing signal

INTERACTION

- Select a virtual patient
- Advance sensor time
- Change meal and activity
- Compare model scenarios

Two explicit data streams

No real patient records, identifiers, or clinical signals are included.

STREAM A · SYNTHETIC EHR

patients.json
6 virtual profiles with age, BMI, HbA1c, baseline glucose, synthetic diagnoses, lab timeline and twin parameters.

STREAM B · TIME SERIES

observations.csv
24 hours per virtual patient at five-minute intervals: glucose, meal events/carbs, steps, sleep and activity.

Model: four Random Forest regressors

Training rows are generated from a transparent stylized synthetic response process.

15 min
Near-term trajectory

30 min
Meal response direction

60 min
Short-horizon forecast

120 min
Excursion window

Features: current/baseline glucose, HbA1c, BMI, age, carbs, minutes since meal, glucose slope, steps, sleep and simulated person-specific parameters.

Split by virtual-patient ID to reduce leakage between generated train/test partitions.

State update loop

01 Load EHR
Patient baseline + history

02 Ingest
Latest sensor observations

03 Refresh
Current twin state + trend

04 Forecast
Multi-horizon prediction

05 Compare
Hypothetical scenario inputs

06 Repeat
Append next simulated sample

Implementation architecture

React + Vite

Recharts dashboard

FastAPI REST

Schemas + in-memory stream

scikit-learn

Random Forest × 4 horizons

Storage: version-controlled synthetic JSON/CSV.
API: patient read, observation read, simulate, forecast, scenario, metrics, reset.
Runtime: local development server; no cloud services or patient database required.

Doctor-facing dashboard concepts

Dashboard summarizes input context so the model is inspectable, not a black box.

Twin state	Observed history + forecast curve
Profile	Synthetic EHR and lab timeline
Scenario studio	Edit meal carbs, steps and sleep
Transparency	Held-out synthetic MAE/RMSE + limitations

Evaluation approach and interpretation

MEASURES

- MAE (mg/dL)
- RMSE (mg/dL)
- Held-out synthetic patient groups
- Tree ensemble spread reported as heuristic only

WHAT RESULTS MEAN

Metrics assess fit to generated trajectories only. Same-process synthetic validation can demonstrate pipeline mechanics, not generalization to real physiology or clinical utility.

Safety, limitations and responsible next steps

NOT CLAIMED

- No clinical validation
- No real CGM integration
- No diagnostic or medication advice
- No calibrated medical probabilities
- No evidence of intervention benefit

NEXT VALIDATION STEPS

- Evaluate on appropriately licensed open data
- Use patient-separated external validation
- Assess calibration and subgroup errors
- Clinician oversight, privacy/security controls
- Prospective study and regulatory review

Live demo: end-to-end in five actions

01	**Select patient**	Open GTB-002 and inspect synthetic EHR.
02	**Inspect history**	Read the CGM curve and last sample.
03	**Forecast**	Update 15/30/60/120-minute trajectory.
04	**Simulate**	Advance virtual sensor by five minutes.
05	**What-if**	Change carbs and compare scenarios.

Reproducible, inspectable code submission

INCLUDED

- React/Vite frontend
- FastAPI backend
- JSON + CSV seed data
- Automated endpoint tests
- README + MIT license
- Architecture PDF/SVG + this deck

RUN LOCALLY

Backend: pip install -r requirements.txt
uvicorn app.main:app --reload

Frontend: npm install
npm run dev

From proof-of-concept to responsible research

1 · Data	Use licensed datasets; document provenance and permissions.
2 · Evaluation	Compare with simple baselines on separated patients.
3 · Clinical review	Co-design outcomes and alert thresholds with clinicians.
4 · Security	Authentication, encryption, audit logging, privacy review.
5 · Validation	Prospective validation and relevant regulatory review.

Measured model fit on held-out synthetic profiles

Values from the committed model configuration; not clinical accuracy estimates.

HORIZON	RF MAE	RF RMSE	PERSISTENCE MAE	PERSISTENCE RMSE
+15 min	2.01	2.63	3.94	5.06
+30 min	2.40	3.16	6.98	8.62
+60 min	3.21	4.22	12.67	15.36
+120 min	3.66	4.77	18.69	23.02

Synthetic test groups only · same generator assumptions · no real-world or clinical validation

The main contribution

A demonstrable pipeline that fuses historical and dynamic data, updates a virtual patient, forecasts a specific outcome, and exposes the scenario assumptions — with transparent limits.

Synthetic-data proof-of-concept • not for clinical decisions